

MA511: Computer Programming

LAB 4 Assignment (Due: 14th Aug 2025)

In this lab you will learn the concepts of **coercion**, **casting** and **functions**.

You must upload the code and the output screenshot of the following assignments.

1. Write a C program that takes as input the time in hours, minutes and seconds. The program should convert the time to seconds
Extend the program to calculate the percentage of day this time represents.
2. Write a program to compute the Euler's number for a given n. The Euler's number is given as

$$e = \sum_{k=0}^n \frac{1}{k!} = \frac{1}{0!} + \frac{1}{1!} + \cdots + \frac{1}{n!}$$

Use function to compute the factorials.

3. Write a program that calculates the HCF and LCM of two given numbers. Use functions to compute the HCF and LCM.
4. Write a C program that accepts two numbers from the user: an integer (iVal) and a floating point number (fVal). The program will ask the user to predict the result type and predicted value. The program will display the Predicted type & value (as entered by the student) and actual type & value (computed by the program). Finally the program will print whether the prediction was correct or not.

The expressions to predict are (a) int + float and (b) int / float.

Sample Output:

Enter an integer: 10

Enter a float: 2.5

Predict for: int + float

Enter predicted type code (1=int, 2=float, 3=double): 2

Enter predicted value: 12.5

Actual type: 2 (float), Actual value: 12.5000 -> MATCH

Predict for: int / float

Enter predicted type code (1=int, 2=float, 3=double): 2

Enter predicted value: 4

Actual type: 2 (float), Actual value: 4.0000 -> MATCH

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